PCB Inspection Report

Timestamp	2026-04-05 14:50:45
PCB Name	Test 1
Source	upload
Status	DEFECTIVE
Risk Level	HIGH
Failure Probability	70.06%

Detected Defects

#	Label	Confidence	BBox
1	Missing_hole	0.7954	[1192, 280, 1214, 301]
2	Missing_hole	0.6899	[802, 55, 848, 100]
3	Missing_hole	0.6693	[705, 55, 745, 98]
4	Missing_hole	0.649	[532, 274, 553, 297]
5	Missing_hole	0.5945	[108, 198, 131, 222]
6	Missing_hole	0.3384	[598, 56, 639, 98]
7	Missing_hole	0.3038	[954, 704, 973, 722]

Explanation

1) Inspection Status
- Overall status: DEFECTIVE
- Risk level: HIGH
- Predicted failure probability: 70.06%
2) Defect Assessment
- Defect 1: Missing_hole | model_conf=0.80 | fused_conf=0.80 | bbox=[1192, 280, 1214, 301]
- Defect 2: Missing_hole | model_conf=0.69 | fused_conf=0.69 | bbox=[802, 55, 848, 100]
- Defect 3: Missing_hole | model_conf=0.67 | fused_conf=0.67 | bbox=[705, 55, 745, 98]
- Defect 4: Missing_hole | model_conf=0.65 | fused_conf=0.65 | bbox=[532, 274, 553, 297]
- Defect 5: Missing_hole | model_conf=0.59 | fused_conf=0.59 | bbox=[108, 198, 131, 222]
- Defect 6: Missing_hole | model_conf=0.34 | fused_conf=0.34 | bbox=[598, 56, 639, 98]
- Defect 7: Missing_hole | model_conf=0.30 | fused_conf=0.30 | bbox=[954, 704, 973, 722]
3) Operational Risk Impact
- High probability of field failure if unreworked defects remain.
4) Recommended Actions
- Rework defects by label priority (short/open/missing first).
- Reinspect repaired regions under magnification and AOI.
- Perform continuity + isolation checks on affected paths.
5) Release Decision Guidance
- Hold production release until full rework and electrical validation are complete.
- Full rework includes repairing all detected defects with appropriate techniques (soldering, component replacement, trace repair).

- Electrical validation encompasses continuity testing, isolation checks, and functional testing under operational conditions.
- Release without validation risks field failures, warranty claims, and potential safety issues in critical applications.
- Validation timeline should include thermal cycling and stress testing to ensure long-term reliability.
- Gemini unavailable reason: GEMINI_API_KEY is missing or placeholder.
- This fallback explanation is generated without Gemini response.

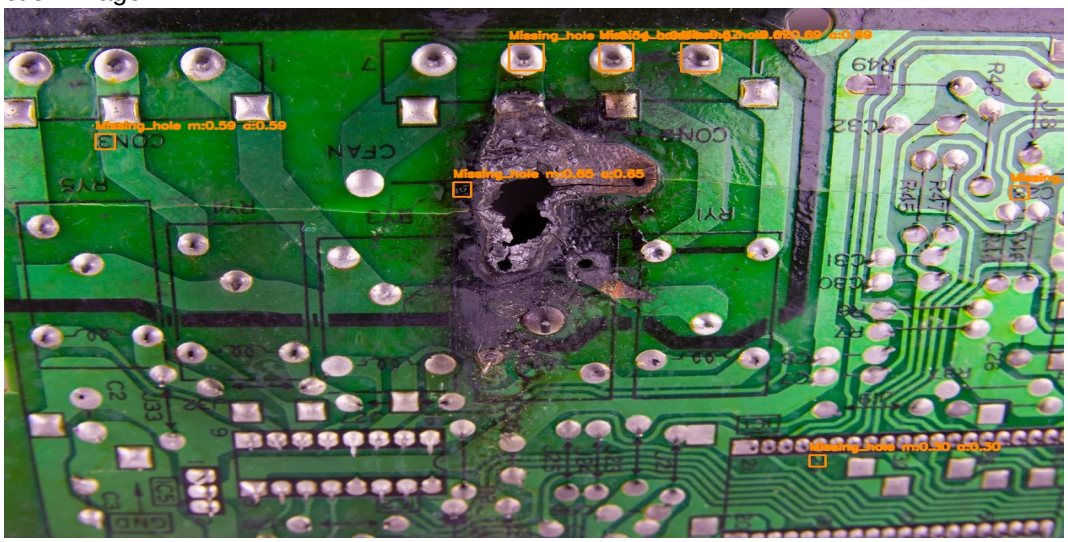
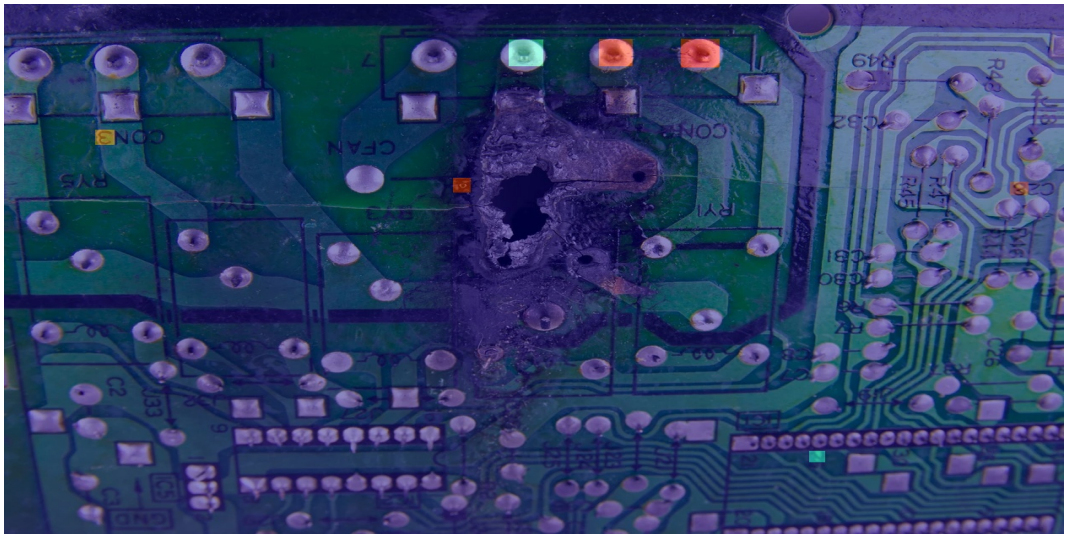
Repair Suggestions

- Missing_hole: Replace the missing component and verify orientation and BOM value before reflow.

Inspection Images

Original Image

Detection Image



Heatmap Overlay